Energy resources and energy transfers

Mark scheme and worked solutions for 25 questions: 8 multiple choice, 17 structured.

1 · Part 1

Paper 1 & 2 | Units | [1]

Answer: B

Explanation

Work done is a transfer of energy, so it is measured in joules (J). A is wrong because the watt is the unit of power, not energy. C is wrong because power is measured in watts (J/s), not joules. D is wrong because force is measured in newtons (N); the kilogram is the unit of mass.

2 · Part 1

Paper 1 & 2 | Energy transfers | [1]

Answer: D

Worked solution

Ask "where did the energy come from?" The only store that was filled up before release is the spring's elastic store, so that is the store that empties as the toy speeds up.

3 · Part 1

Paper 1 & 2 | Energy transfers | [1]

Answer: A

Worked solution

Efficiency uses the useful energy, not the wasted energy. Subtract the waste first: 2500 J − 750 J = 1750 J useful. Then efficiency = useful energy output / total energy supplied × 100% = 1750 / 2500 × 100% = 70%. Check: 70% useful + 30% wasted = 100%. An efficiency can never be more than 100%, so C is impossible.

4 · Part 1

Paper 1 & 2 | Energy transfers | [1]

Answer: C

Explanation

Radiation (infrared) is an electromagnetic wave and needs no particles, so it can travel through a vacuum. A and B are wrong because conduction and convection both need particles: conduction passes energy between neighbouring particles and convection needs a fluid that can flow. D is wrong for the same reason.

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5 · Part 1 Paper 1 & 2 Work and power [1]

Answer: C

Worked solution

Convert 2.0 km to 2000 m first, because the joule is a newton × metre. Then work done = 300 × 2000 = 600 000 J. Check: 600 000 J / 300 N = 2000 m, which is 2.0 km.

6 · Part 1 Paper 1 & 2 Work and power [1]

Answer: A

Worked solution

Use conservation of energy: as the ball rises it slows down (kinetic store decreases) and gets higher (gravitational store increases). With nothing wasted, every joule lost from one store goes into the other.

7 · Part 1 Paper 1 & 2 Work and power [1]

Answer: D

Worked solution

Power is the rate of doing work, so divide work done by time, with the time in seconds. 1.5 × 60 = 90 s, and 36 000 / 90 = 400 W. Check: 400 W × 90 s = 36 000 J.

8 · Part 1 Paper 2 only Energy resources and electricity generation [1]

Answer: B

Explanation

Water held high behind the dam has energy in its gravitational store. As it falls, this is transferred to the kinetic store of the moving water and then the turbine, and the generator transfers the energy electrically. A is wrong because the water is not hot; no thermal store drives the turbine. C is wrong because water is not a fuel and no chemical store is used. D is wrong because the dam does not store elastic energy and the output is transferred electrically, not by heating.

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9 · Part 1 Paper 1 & 2 Energy transfers [4]

Mark scheme

(a) **[1]**

- elastic (potential) store (of the band) (1)

ALLOW strain energy store

(b) **[2]**

- energy transferred (mechanically) from the elastic store of the band to the kinetic store of the ball (1)
- (as the ball rises) energy transferred from the kinetic store of the ball to its gravitational (potential) store (1)

IGNORE reference to small amounts transferred to the thermal store of the surroundings by air resistance

REJECT "chemical store" of the band

(c) **[1]**

- energy cannot be created or destroyed, only transferred (from one store to another) (1)

ALLOW the total energy of a (closed) system stays the same

Worked solution

(a) Stretching the band stores energy in its elastic store. (b) When the band is released it pushes the ball, transferring energy mechanically into the ball's kinetic store. As the ball rises it slows down: its kinetic store empties and its gravitational store fills. At the highest point the ball is momentarily still, so (ignoring air resistance) all the energy is in the gravitational store. (c) The principle of conservation of energy says energy is never created or destroyed; it is only transferred between stores, so the total stays the same.

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10 · Part 1 Paper 1 & 2 Energy transfers [8]

Mark scheme

(a) **[2]**

- efficiency = (30 / 600) × 100% (1)
- = 5% (1)

ALLOW 0.05

(b) **[2]**

- 80 − 32 = 48 J (1)
- thermal store of the surroundings (1)

ALLOW thermal store of the lamp OR heating the surroundings

(c) **[2]**

- width = (30 / 600) × 12 OR 0.05 × 12 (1)
- = 0.60 cm (1)

ALLOW 0.6 cm; ALLOW ECF from (a)

(d) **[2]**

- the LED gives about the same (or slightly more) light (32 J compared with 30 J) for much less energy supplied (80 J compared with 600 J) (1)
- the LED is more efficient (40% compared with 5%) so it wastes much less energy heating the surroundings (1)

ALLOW the LED wastes 48 J in 10 s but the filament lamp wastes 570 J

Worked solution

(a) Efficiency = useful energy output / total energy supplied × 100% = 30 / 600 × 100% = 5%. (b) Conservation of energy: energy wasted = 80 − 32 = 48 J, which ends up heating the lamp and the room (the thermal store of the surroundings). (c) In a Sankey diagram, arrow width is proportional to energy. The light arrow is 30 / 600 = 0.05 of the input, so its width is 0.05 × 12 cm = 0.60 cm. (d) The LED gives a similar amount of light but needs only 80 J instead of 600 J every 10 s, because its efficiency is 32 / 80 × 100% = 40% rather than 5%. Less energy is wasted as heat, so less needs to be supplied for the same lighting.

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11 · Part 1 Paper 1 & 2 Energy transfers [6]

Mark scheme

(a) **[1]**

- conduction (1)

(b) **[2]**

- particles (atoms) in the hotter part of the metal vibrate more (gain kinetic energy) (1)
- they pass energy on to neighbouring particles (by collisions) OR free electrons gain energy and move through the metal carrying energy to cooler parts (1)

(c) **[3]**

- soup near the base is heated, expands and becomes less dense (1)
- the warmer soup rises and cooler, denser soup sinks to take its place (1)
- a convection current is set up that circulates (all) the soup, so it is all heated (1)

REJECT "heat rises" with no reference to the soup (fluid) moving

Worked solution

(a) and (b) Metals are good conductors. In the base, the particles nearest the hob vibrate more strongly and pass energy to their neighbours, and metals also have free electrons that move through the metal carrying energy quickly to the cooler side. (c) Soup is a liquid, so convection can happen. The soup at the bottom is warmed, expands and becomes less dense, so it rises. Cooler, denser soup sinks to the bottom to be heated in turn. This circulation (a convection current) carries energy through all of the soup.

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12 · Part 1 Paper 1 & 2 Energy transfers [7]

Mark scheme

(a) **[1]**

- (infrared) radiation (1)

> **ALLOW** electromagnetic waves OR light and infrared

(b) **[4]**

- air above the land is heated by the (warmer) land (1)
- this air expands and becomes less dense (1)
- the warm, less dense air rises (1)
- cooler, denser air from above the sea moves in (towards the land) to replace it, forming a convection current (1)

> **REJECT** "the sea breeze is caused by cold rising" OR "heat rises" with no reference to density

(c) **[2]**

- from the land towards the sea (1)
- the air above the (now warmer) sea is heated, becomes less dense and rises, so cooler air from above the land moves out to replace it (1)
- the reason mark depends on a correct direction

Worked solution

(a) The Sun's energy reaches Earth by radiation, the only method that works through space. (b) The warm land heats the air just above it. That air expands, becomes less dense and rises. Cooler, denser air from over the sea moves in to take the place of the rising air, so wind blows from sea to land. Higher up, the air moves back out to sea and sinks, completing a convection current. (c) At night everything reverses: the sea is now the warmer surface, so air rises above the sea and cooler air flows from the land out to sea.

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13 · Part 1 Paper 1 & 2 Energy transfers [7]

Mark scheme

(a) **[1]**

- the type (colour OR shininess) of the surface (1)

(b) **[2]**

- any two from (1 each):
- distance from the heater
- size OR area of the plate
- thickness OR type of metal of the plate
- starting temperature of the plates
- the same heater OR heater power

REJECT the time taken (this is the dependent variable)

(c) **[2]**

- matt black (plate X) (1)
- its temperature rose by 10 °C in a shorter time (45 s compared with 160 s), so it absorbed infrared energy faster (1)

(d) **[2]**

- plate X (matt black) (1)
- matt black surfaces are better emitters of infrared radiation (than shiny silver surfaces) (1)

ALLOW shiny silver is a poor emitter

- the explanation mark depends on X being chosen

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Worked solution

(a) The variable the student chooses to change is the type of surface. (b) For a fair test, everything else that could affect the result must stay the same, for example each plate's distance from the heater (20 cm), its size and thickness, and its starting temperature. (c) The matt black plate warmed by 10 °C in 45 s while the silver plate took 160 s, so the black plate absorbed infrared energy more than three times as fast. (d) Good absorbers are also good emitters. Matt black surfaces emit infrared radiation faster than shiny silver ones, so plate X loses energy and cools faster.

14 · Part 1 Paper 1 & 2 Energy transfers [6]

Mark scheme

(a) **[1]**

- the time taken for the pin to fall off (1)

(b) **[2]**

- copper (1)
- its pin fell off in the shortest time OR its wax melted first (1)
- the reason mark depends on copper

(c) **[2]**

- to make it a fair test so that only the material changes (1)
- length and thickness affect how quickly energy reaches the wax (for example a longer rod takes longer) (1)

(d) **[1]**

- any one safety precaution, e.g.:
- handle the hot water carefully (to avoid scalds) OR keep the beaker on a stable heat-proof mat (1)

ALLOW do not touch the rods (they become hot) OR wear eye protection

IGNORE "wear a lab coat" alone

Worked solution

(a) The dependent variable is what the student measures: the time for each pin to fall. (b) The better the conductor, the faster energy travels along the rod and melts the wax. Copper's pin fell first (35 s), so copper is the best conductor. Glass is a poor conductor (an insulator), so its wax had not melted after 600 s. (c) A longer or thinner rod would take longer to conduct enough energy to the wax, so differences in size would make the comparison unfair. (d) The water is hot and the rods become hot, so take care not to be burned or scalded.

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15 · Part 1 Paper 1 & 2 Energy transfers [6]

Mark scheme

(a) **[2]**

- (trapped) air is a poor conductor (insulator), so conduction through the window is reduced (1)
- the gap is thin so the air cannot circulate easily, reducing convection (1)

ALLOW if gas or vacuum in the gap is described correctly

(b) **[2]**

- trapped air in the pockets is a poor conductor, so conduction is reduced (1)
- the air cannot move (circulate), so convection currents are prevented OR reduced (1)

(c) **[2]**

- payback times calculated: loft 600 / 200 = 3 years, cavity wall 1500 / 300 = 5 years, double glazing 4800 / 160 = 30 years (1)
- loft insulation (1)
- the method mark needs at least two payback times correct; "loft insulation" with no working scores 0

Worked solution

(a) Air is a very poor conductor of thermal energy. Because the layer is thin, the air cannot circulate much, so convection between the panes is small too. (b) Fibreglass traps air in tiny pockets. The trapped air conducts poorly and cannot flow, so convection currents cannot form. (c) Payback time = cost to install / saving each year. Loft insulation: 600 / 200 = 3 years. Cavity wall: 1500 / 300 = 5 years. Double glazing: 4800 / 160 = 30 years. Loft insulation pays for itself fastest.

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16 · Part 1 Paper 1 & 2 Energy transfers [6]

Mark scheme

(a) **[1]**

- a vacuum contains no particles (to pass on vibrations or to flow) (1)

(b) **[5]**

- indicative points, 1 mark each, max 5:
- the vacuum stops conduction between the inner and outer walls
- the vacuum stops convection between the walls
- the silvered surfaces reflect infrared radiation (back towards the drink)
- the silvered surfaces are poor emitters of infrared radiation
- the plastic stopper is a poor conductor, so it reduces conduction out through the top
- the stopper stops warm air rising out of the flask, reducing convection (and evaporation)
- the insulating supports (and plastic case) are poor conductors, reducing conduction to the outer case
- at least one point must refer to radiation for full marks

Worked solution

(a) Conduction and convection both need particles: conduction passes energy between neighbouring particles, and convection needs a fluid to flow. A vacuum has no particles, so neither can happen. (b) The flask attacks all three methods. The vacuum between the walls blocks conduction and convection through the sides. Radiation can cross a vacuum, so the walls are silvered: shiny surfaces are poor emitters and good reflectors of infrared, so little radiation leaves the hot inner wall and much of it is reflected back. The plastic stopper conducts poorly and seals the top, stopping hot air rising out by convection. The insulating supports touch the glass only in small places and conduct poorly, so little energy is conducted to the outer case.

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17 · Part 1 Paper 1 & 2 Work and power [7]

Mark scheme

(a) **[2]**

- GPE = 55 × 10 × 3.6 (1)
- = 1980 J (1)

ALLOW 2000 J

(b) **[1]**

- 1980 J (1)

ALLOW ECF from (a)

(c) **[2]**

- power = 1980 / 4.5 (1)
- = 440 W (1)

ALLOW ECF from (b)

(d) **[2]**

- the second student does the same work OR transfers the same energy (same mass and same height) (1)
- in less time, so the rate of transfer of energy is greater: greater power (1)

ALLOW calculation of 660 W for the second student as part of the explanation

Worked solution

(a) Gravitational potential energy = mass × gravitational field strength × height = 55 × 10 × 3.6 = 1980 J. (b) Work done is equal to energy transferred, so the work done against gravity is 1980 J. (c) Power = work done / time taken = 1980 / 4.5 = 440 W. Check: 440 W × 4.5 s = 1980 J. (d) Power is the rate of doing work. The second student does the same 1980 J of work, but in 3.0 s instead of 4.5 s, so the power is greater (1980 / 3.0 = 660 W).

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18 · Part 1 Paper 1 & 2 Work and power [8]

Mark scheme

(a) **[2]**

- KE = ½ × 80 × 6.0^2 (1)
- = 1440 J (1)

ALLOW 1400 J

(b) **[2]**

- KE = ½ × 80 × 12^2 = 5760 J (1)

ALLOW 5800 J

- (increased by a factor of) 4 (1)

ALLOW ECF for the factor from the two calculated values

(c) **[3]**

- work done by the brakes = kinetic energy lost = 1440 J (1)
- 1440 = F × 18 (1)
- F = 80 N (1)

ALLOW ECF from (a)

(d) **[1]**

- thermal store of the brakes (and wheels) OR of the surroundings (1)

ALLOW heating the brakes

Worked solution

(a) KE = ½ × m × v^2 = 0.5 × 80 × 36 = 1440 J. (b) At 12 m/s, KE = 0.5 × 80 × 144 = 5760 J. The speed doubled, but KE depends on speed squared, so KE increases by 2^2 = 4 times (5760 / 1440 = 4). (c) To stop, the brakes must do work equal to all the kinetic energy, 1440 J. Work done = force × distance, so F = 1440 / 18 = 80 N. Check: 80 N × 18 m = 1440 J. (d) Friction in the brakes transfers the energy by heating to the thermal store of the brakes, which then heats the surroundings.

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19 · Part 1 Paper 1 & 2 Work and power [7]

Mark scheme

(a) **[2]**

- KE = ½ × 60 × 9.0^2 (1)
- = 2430 J (≈ 2400 J) (1)
- the value 2430 J must be seen to score the second mark

(b) **[3]**

- GPE gained = KE OR 2430 = 60 × 10 × h (1)
- h = 2430 / 600 (1)
- = 4.05 m (1)

ALLOW 4.0 m (using 2400 J) OR 4.1 m

(c) **[2]**

- any two from (1 each):
- the pole vaulter is still moving (horizontally) at the top, so some energy remains in the kinetic store
- some energy is transferred to the thermal store of the surroundings (air resistance, or heating in the pole and in the body)
- some energy is transferred as sound OR some energy stays in the elastic store of the pole

REJECT "energy is lost" with no store or transfer named

Worked solution

(a) KE = ½ × m × v^2 = 0.5 × 60 × 81 = 2430 J, which is about 2400 J. (b) If all the kinetic energy becomes gravitational potential energy, then m × g × h = 2430 J, so h = 2430 / (60 × 10) = 4.05 m. Check: 60 × 10 × 4.05 = 2430 J. (c) Not all of the kinetic energy reaches the gravitational store. The vaulter must still be moving forward at the top to clear the bar, so some energy stays in the kinetic store, and some is transferred to the surroundings by heating and as sound.

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20 · Part 1 Paper 1 & 2 Work and power [5]

Mark scheme

(a) **[2]**

- work done = 150 × 12 (1)
- = 1800 J (1)

(b) **[1]**

- thermal store of the crate and floor OR of the surroundings (because of friction) (1)

REJECT kinetic store

(c) **[2]**

- work done = force × distance moved in the direction of the force (1)
- the cupboard does not move, so the distance is zero and the work done is zero OR no energy is transferred to the cupboard (1)

Worked solution

(a) Work done = force × distance moved = 150 N × 12 m = 1800 J. (b) The crate moves at constant speed, so its kinetic store does not change. The 1800 J is transferred, by friction between the crate and the floor, to the thermal store of the crate, floor and surroundings. (c) Work is only done when a force moves something in the direction of the force. The cupboard does not move, so distance moved = 0 and work done = 150 × 0 = 0 J. The worker gets tired because their muscles transfer energy internally, but none of it goes to the cupboard.

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21 · Part 1 Paper 1 & 2 Work and power [8]

Mark scheme

(a) **[2]**

- work done = 600 × 500 (1)
- = 300 000 J (1)

ALLOW 3.0×10^5 J

(b) **[2]**

- power = 300 000 / 20 (1)
- = 15 000 W (1)

ALLOW ECF from (a)

(c) **[2]**

- the speed is constant, so there is no increase in kinetic energy (1)
- the work done is against resistive forces (air resistance and friction) and is transferred to the thermal store of the surroundings (1)

(d) **[2]**

- efficiency = (15 000 / 60 000) × 100% (1)
- = 25% (1)

ALLOW 0.25; ALLOW ECF from (b)

Worked solution

(a) Work done = force × distance moved = 600 N × 500 m = 300 000 J. (b) Power = work done / time taken = 300 000 / 20 = 15 000 W. (c) The car moves at a constant speed, so its kinetic store stays the same. The engine's work is done against air resistance and friction, which transfer the energy by heating to the surroundings. (d) Efficiency = useful power output / total power input × 100% = 15 000 / 60 000 × 100% = 25%. Check: three quarters of the fuel's energy (45 000 W) is wasted, mainly heating the engine and surroundings.

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22 · Part 1 Paper 1 & 2 Work and power [8]

Mark scheme

(a) **[2]**

- GPE = 4.0 × 10 × 1.5 (1)
- = 60 J (1)

(b) **[2]**

- KE = ½ × 4.0 × 4.0^2 (1)
- = 32 J (1)

(c) **[2]**

- (by conservation of energy) GPE lost = KE gained + work done against friction OR work done against friction = 60 − 32 (1)
- = 28 J (1)

ALLOW ECF from (a) and (b)

(d) **[2]**

- 28 = F × 3.0 (1)
- F = 9.3 N (1)

ALLOW 9.33 N; ALLOW ECF from (c)

- using the vertical height (1.5 m) instead of the ramp length gives 19 N: award the first mark only

Worked solution

(a) GPE lost = m × g × h = 4.0 × 10 × 1.5 = 60 J. (b) KE = ½ × m × v^2 = 0.5 × 4.0 × 16 = 32 J. (c) Energy is conserved. Of the 60 J lost from the gravitational store, only 32 J reached the kinetic store, so the rest, 60 − 32 = 28 J, was transferred by work done against friction to the thermal store of the box, ramp and surroundings. (d) The friction acts along the ramp over its full length of 3.0 m, so work done = force × distance gives F = 28 / 3.0 = 9.3 N. Check: 9.33 × 3.0 = 28 J.

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23 · Part 1 Paper 2 only Energy resources and electricity generation [6]

Mark scheme

(a) **[4]**

- indicative points, 1 mark each, max 4:
- energy is in the chemical store of the gas (fuel)
- burning the gas transfers energy by heating to the thermal store of the water, turning it into steam
- the (high-pressure) steam turns the turbine: energy is transferred to the kinetic store of the turbine
- the turbine turns the generator, which transfers the energy electrically (to the grid)
- some energy is wasted, transferred by heating to the thermal store of the surroundings

(b) **[1]**

- energy transferred by (light) radiation from the Sun is transferred electrically by the cell (1)

ALLOW light energy → electrical energy

(c) **[1]**

- nuclear store (1)

Worked solution

(a) Natural gas has energy in its chemical store. Burning it heats water in a boiler, transferring energy to the thermal store of the water and producing high-pressure steam. The steam pushes on the turbine blades, so the turbine gains energy in its kinetic store. The spinning turbine turns a generator, which transfers energy electrically to the grid. Much of the energy is wasted heating the surroundings, for example in the cooling towers. (b) A solar cell takes energy transferred by light radiation from the Sun and transfers it electrically, with no moving parts. (c) Nuclear fuel such as uranium has energy in its nuclear store; fission releases it to heat water, and the rest of the station works like a fossil-fuel station.

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24 · Part 1 Paper 2 only Energy resources and electricity generation [7]

Mark scheme

(a) **[3]**

- indicative points, 1 mark each, max 3:
- energy comes from the thermal store of the hot rocks underground
- (cold) water is pumped down to the hot rocks and is heated, becoming steam OR hot water
- the steam (or hot water) rises and turns a turbine (kinetic store)
- the turbine drives a generator, which transfers energy electrically

(b) **[2]**

- energy from the kinetic store of the moving air (wind) is transferred to the kinetic store of the turbine (blades) (1)
- the turbine turns a generator, which transfers energy electrically (1)

(c) **[1]**

- geothermal power is reliable OR available all the time, because it does not depend on the weather (wind speed) (1)

ALLOW geothermal takes up less land area for the same output

IGNORE "renewable" (both are renewable)

(d) **[1]**

- any one, e.g.: only possible in a few places (where hot rocks are near the surface) (1)

ALLOW drilling deep is expensive OR harmful gases may be released from underground

Worked solution

(a) The rocks deep underground have a large thermal store. Water is pumped down, heated by the rocks and returns as steam or very hot water. This turns turbines, which drive generators that transfer energy electrically. (b) Moving air has energy in its kinetic store. It pushes the turbine blades round, transferring energy to their kinetic store, and the generator then transfers it electrically. (c) The wind does not always blow, but the hot rocks are always hot, so a geothermal station can supply electricity all the time. (d) Geothermal power needs hot rocks close enough to the surface, which only happens in certain regions, and the deep drilling is costly.

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25 · Part 1 Paper 2 only Energy resources and electricity generation [7]

Mark scheme

(a) **[1]**

- a resource that is replaced (naturally) as it is used, so it will not run out (1)

(b) **[6]**

- indicative points, 1 mark each, max 6:
- nuclear advantage: no carbon dioxide (greenhouse gases) produced while generating electricity
- nuclear advantage: reliable, steady output day and night, whatever the weather
- nuclear advantage: a small mass of fuel releases a very large amount of energy OR the station needs little land
- nuclear disadvantage: produces radioactive waste that must be stored safely for a very long time
- nuclear disadvantage: risk of an accident releasing radioactive material
- nuclear disadvantage: very expensive to build and to decommission OR takes a long time to build
- nuclear disadvantage: uranium is a non-renewable resource
- solar advantage: renewable, energy resource will not run out
- solar advantage: no fuel costs OR low running costs
- solar advantage: no carbon dioxide produced while generating electricity
- solar disadvantage: no output at night and less in cloudy weather, so it is unreliable OR needs backup or storage
- solar disadvantage: needs a very large area of land for the same output
- max 4 if only one replacement is discussed
- max 4 if only advantages or only disadvantages are given

Worked solution

(a) A renewable resource, such as sunlight or wind, is replaced naturally and will not run out. (b) A good answer weighs both options. Nuclear power: it releases no carbon dioxide while running and gives a large, steady output whatever the weather, from a small mass of fuel on a small site. But it produces radioactive waste that stays dangerous for thousands of years, there is a small risk of a serious accident, it is very expensive and slow to build and decommission, and uranium is non-renewable. Solar farm: sunlight is renewable and free, running costs are low and no carbon dioxide is produced while generating. But it produces nothing at night and less on cloudy days, so it needs backup or energy storage, and it needs a huge area of land to match a power station's output. The choice depends on whether reliability or avoiding radioactive waste matters more.

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